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Studierichting: Informatica
Oefeningenreeks 4A nr. 14.13

$$
\begin{aligned}
I &= \int \frac{x \ln x \cos x - \sin x}{x(\ln x)^2} dx \\
&= \int \frac{x \ln x \cos x}{x \ln x \ln x} dx - \int \frac{\sin x}{x \ln x \ln x} dx \\
&= \int \cos x \frac{1}{\ln x} dx - \int \frac{\sin x}{x \ln x \ln x} dx \\
&= \int \cos x \frac{1}{\ln x} dx - \int \sin x \frac{1}{x \ln x \ln x} dx
\end{aligned}
$$

Partiële integratie van de 1ste integraal: $\begin{cases} u = & \dfrac{1}{\ln x} \\ dv = & \cos x dx \end{cases} \Rightarrow \begin{cases} du = & \dfrac{-1}{x \ln x \ln x} dx \\ v = & \sin x \end{cases}$

$$
\begin{aligned}
I &= \frac{\sin x}{\ln x} - \int \sin x \frac{-1}{x \ln x \ln x} dx - \int \sin x \frac{1}{x \ln x \ln x} dx \\
&= \frac{\sin x}{\ln x} + \cancel{\int \sin x \frac{1}{x \ln x \ln x} dx} - \cancel{\int \sin x \frac{1}{x \ln x \ln x} dx} \\
&= \frac{\sin x}{\ln x} + c
\end{aligned}
$$

References